

Wilson identification and a homogeneous update: a Round27 research addendum

Research workbench draft for author review

22 September 2026

Abstract

This addendum extends the unchanged first manuscript draft through a selected historical-method panel and three separately reviewed mathematical investigations. The readout analysis characterizes the exact parameter fibers of specified Wilson endpoint surrogates and identifies the calibration information absent from a universal limiting curve. A subsequent finite-profile study compares actual bounded Wilson multiplication observables, retaining their full stationary, spatial, averaging and arithmetic errors. A separate homogeneous-model study addresses one nonlinear correction with explicit support-weight bookkeeping. The source survey covers selected historical, mystical, technical, patent, institutional and discussion material; its role is to propose discriminating questions, not to supply a new physical premise. We record short derivations, reproducible calculations, limitations and the next missing estimates. None of these results constructs four-dimensional continuum Yang–Mills theory or proves its physical mass gap.

Status and scope. This is a research addendum, not an arXiv submission. Human authorship and affiliations remain to be supplied. Separately authored model-agent checks are not external peer review or formal verification. The marker ** denotes a contribution derived or implemented within this workbench; scientific priority is unverified.

1 Inherited question and research design

The unchanged Draft01 compendium records the surviving history through Round26. It distinguishes a finite Gauss-invariant $SU(2)$ graph, a homogeneous local-stability model and a canonical summable full-link family. Its 293-coordinate heat calculation, complete-support source estimates and Wilson endpoint formulas have different hypotheses. A shared parameter name does not identify their physical clocks, generators, states or domains. Round27 therefore begins with an information question: which independent parameters do the available readouts actually identify?

The panel comprised contemporary model-agent research lenses informed by selected Newton, Tesla, Jung/Pauli, Penrose and Feynman sources. They neither impersonate those people nor claim their endorsement. The advisor chose a contract; separate forward and reverse researchers derived and implemented it without reading the other’s current solution before freezing; a separate skeptic tested scope and counterexamples. The mathematical work used the paired directions where useful. The geometry of a double spiral was an organizing analogy, not a proof premise.

The first selection, AI1, analyzes exact readout fibers. Its review motivates AI2’s new physical probes. Only after AI2’s review is the third contract selected, addressing the homogeneous nonlinear-update obstruction. This sequencing matters: the later question must respond to an observed limitation rather than retroactively redefining the first target. Literature searches, replays, repairs and skill tests do not count as additional physics loops.

Fixed units. Throughout the canonical identification studies, $\alpha > 0$ is an energy, $\hbar > 0$ is a fixed action reference, and $E_* > 0$ is a separately fixed physical energy scale. The dimensionless parameters satisfy $0 < \eta, q < 1$. Neither q nor $1 - q$ is asserted to be a continuum lattice spacing. The homogeneous study has its own declared dimensionless coupling and support weights. A support weight is proof bookkeeping, not a new force or physical constant.

2 What the sources contribute

The five inventories contain 55 entries at 54 distinct URLs, consulted on 22 September 2026. They retain exact URLs, reading depth and explicit nonclaims. This is a selected survey, not exhaustive access to the works of five researchers or a complete state-of-the-art or priority search. An abstract is not a full-paper audit; a catalogue is not a read manuscript. The mathematical premises come from the frozen repository contracts and inherited reports, not from occult texts, public anecdotes or institutional sponsorship.

Lens	Selected material and documented reading boundary	Question transferred to this work
Newton	<i>De motu, Opticks</i> Query 31, General Scholium; selected alchemical operations and residue notes; Emerald Tablet transcription and theological imagery dictionary.	Does inverse reconstruction identify a unique cause? Are all residuals and support costs retained?
Tesla	Powered transmission patent; one Colorado Springs tuning entry; Vivekananda letter; public custody rumor traced to an official account and archive catalogue. The large archive scan was not read.	What are the input, load, actual observable, clock and independent calibration?
Jung/Pauli	Publisher-hosted alchemy and synchronicity openings; correspondence context, with an inaccessible scan disclosed; two purposively sampled discussions; institutional remote-viewing history.	Which observation rule distinguishes a pattern from an indistinguishable alternative?
Penrose	Spin networks, first two pages of handwritten direction notes, twistor abstract; separately scoped gravitational-reduction proposal.	Do candidate states have a nondegenerate Gram matrix, and does a proposed map apply to this dimension and model?
Feynman	Methodological writing; path-integral and simulation meta-data/abstracts; qualitative $2 + 1$ dimensional Yang–Mills abstract.	What measured quantity and countermodel discriminate a claim? What does the simulation actually construct?

Table 1: Historical sources generate operational questions. They add no Hamiltonian term or mass-gap premise. Full reading limits and links are in the expert inventories.

Newton’s speculative subtle-spirit passage acknowledges inadequate experiments for a demonstrated law. His copied Hermetic text is not a modern scale-matching theorem. Tesla’s patent specifies a powered mechanism; its grant establishes neither operational performance nor quantum-field implications. Vivekananda’s religious interpretation records an encounter and an anticipated demonstration, not a completed Tesla derivation. The custody rumor review separates a Reddit allegation, the FBI’s institutional account and a catalogue whose large scan was inaccessible. These distinctions prevent an attractive story from silently becoming a physical assumption.

Jung’s selected symbolic correspondences motivate declaring observation and selection rules. The reviewed government history establishes that remote-viewing investigations occurred, not a mechanism or a connection to gauge dynamics. Public discussions record competing interpretations; no representative sampling, verified anomalous experience or empirical corpus experiment is claimed

here. Penrose’s warning that locally admissible spin-network labels can still yield a zero-norm network reinforces an existing Gram-matrix gate. His gravitational-reduction proposal changes the physical model if imported into pure Yang–Mills. Feynman’s qualitative $2 + 1$ dimensional paper cannot close a $3 + 1$ dimensional quantum construction.

2.1 Current technical comparisons and their boundaries

The survey follows targeted primary sources through 2026. Newton’s lens inspects local plaquette-generated truncations and selected area-law theorem statements; their group, action and coupling hypotheses must be checked before transfer. Tesla’s scale-setting comparison distinguishes gradient-flow convention, finite-spacing artifacts, topology and volume. Its numerical calibration cannot supply a value of this workbench’s α without a state/observable/generator map. The hardware comparison specifies measured observables and digitization errors on a two-plaquette system; government funding confirms a program, not correctness of a theorem.

The Penrose review inspects the current stochastic three-dimensional Yang–Mills–Higgs result, whose introduction retains construction and invariant-measure limitations. Its dimension, dynamics and target differ from a four-dimensional pure quantum theory. The Feynman review compares statistically bounded time evolution, Hamiltonian step scaling in $2 + 1$ dimensional $U(1)$ and finite-group digitization. These are useful ways to separate approximation, physical scale and continuum errors. They do not establish this workbench’s missing limits. Selected quasi-locality literature similarly motivates the distinction between strong and norm topology, while leaving our declared support-cardinality estimate to be proved.

The survey also preserves skeptical objections to current preprints. One displayed inverse map is invariant under an isotropic shift, so an unrestricted scalar cannot be recovered from it. A gauge-cooling introduction’s claim about neighboring constraints is not adopted: distinct full-link vertex transformations use commuting left and right actions on a shared link. These are narrow source-screening observations, not complete refutations or literature-wide novelty findings. No recent claimed mass-gap solution was imported as an accepted premise.

3 AI1: exact information in the declared readouts^{**}

3.1 Endpoint coordinates and inherited functions

Put $\epsilon = 1 - q$, $\beta = \alpha/\hbar$ and

$$\rho = \eta\epsilon^3, \quad \lambda = \frac{\beta\rho}{84}, \quad T_q(z) = \frac{z}{\beta\eta\epsilon^3}, \quad \theta = \frac{z}{84} = \lambda T_q(z). \quad (1)$$

β and λ have units of inverse time. The inherited endpoint theorem admits $0 < z \leq 10^{-6}$, with α, η fixed and $q \uparrow 1$. Physical observation time then diverges. The elementary plaquette, a prescribed six-cycle X , and a coherent six-cycle pair S have respective endpoint functions

$$\begin{aligned} f_0(\theta) &= 1, \\ f_X(\theta) &= \frac{2}{3} + \frac{1}{4} \cos(2\theta) + \frac{1}{12} \cos(2\sqrt{3}\theta), \\ f_S(\theta) &= \frac{1}{3} + \frac{1}{2} \cos(2\theta) + \frac{1}{6} \cos(2\sqrt{3}\theta) + i \left[\frac{1}{4} \sin(2\theta) + \frac{1}{4\sqrt{3}} \sin(2\sqrt{3}\theta) \right]. \end{aligned} \quad (2)$$

The actual assertion is a demodulated, growing-time limit $e^{iE_j T_q/\hbar} C_{j,q}(T_q) \rightarrow f_j(z/84)$, with $E_0 = 3\alpha$ and $E_X = E_S = 9\alpha/2$. It is not a derivative theorem at fixed physical time. Analytic extension of the finite functions in (2) does not enlarge the physical theorem’s interval.

At fixed scaled coordinate z , the entire limiting readout is independent of α, η, q . It identifies none of them. Recorded physical timestamps supply the composite rate through the protocol; that information is additional to the endpoint ordinates. At fixed physical t , $q \uparrow 1$ instead sends $z = \beta\eta(1 - q)^3 t$ to zero. Confusing these two limits would manufacture an inverse that the data do not provide.

3.2 Exact fibers and conditional identification

Define the physical-time *surrogate*, explicitly distinct from an exact finite- q correlation,

$$D_j(t; \alpha, \eta, q) = f_j(\lambda t), \quad 0 < 84\lambda t \leq 10^{-6}. \quad (3)$$

Substitution proves that all readouts coincide whenever

$$\alpha\eta(1 - q)^3 = \text{constant}.^{**} \quad (4)$$

Conversely, positivity of λ and strict increase of the coherent imaginary part identify this one composite rate. Indeed

$$\frac{d}{d\theta} \text{Im } f_S(\theta) = \frac{1}{2} \cos(2\theta) + \frac{1}{2} \cos(2\sqrt{3}\theta) \geq 1 - 4\theta^2 > 0 \quad (5)$$

on the admitted interval. The inequality uses $\cos x \geq 1 - x^2/2$. The logarithmic sensitivity row is $(1, 1, 3)$ in coordinates $(\log \alpha, \log \eta, \log \epsilon)$; its rank is one. More samples of these same complete curves do not remove the two-dimensional nullspace.

For example, in fixed units the exact triples $(\alpha/E_*, \eta, q) = (1, 1/4, 1/2)$, $(2, 1/8, 1/2)$ and $(1, 2/27, 1/4)$ all have $\alpha\eta(1 - q)^3/E_* = 1/32$. Compensation also persists arbitrarily near $q = 1$. These rational countermodels disprove parameter uniqueness for (3), not equality of their actual finite- q Hamiltonians.

Retain a calibrated carrier:

$$U_0(t) = e^{-3i\beta t}, \quad U_X(t) = e^{-(9/2)i\beta t} f_X(\lambda t), \quad U_S(t) = e^{-(9/2)i\beta t} f_S(\lambda t). \quad (6)$$

On an open physical-time interval, complete complex surrogate curves identify β and λ . For instance $U'_0(0) = -3i\beta$ and $U'_S(0) = i(\lambda - 9\beta/2)$ as analytic identities of these surrogates. Their sensitivity matrix has rows $(1, 0, 0)$ and $(1, 1, 3)$, rank two, leaving the η - q fiber. Known q then gives $\eta = 84\lambda/[\beta(1 - q)^3]$. This conclusion requires calibrated time, action and phase; it is not a measurement already implemented by the endpoint limit.

Sampling also matters. On a uniform grid $t = n\Delta$, the simultaneous change

$$\alpha' = \alpha + \frac{4\pi\hbar m}{3\Delta}, \quad \eta' = \eta\alpha/\alpha', \quad q' = q \quad (7)$$

preserves the slow rate and gives integer- 2π shifts to both carriers, when the transformed parameters remain admissible. A bounded prior frequency range or nonaliased design is needed. Intensity-only data lose the elementary phase entirely. The elementary demodulated probe remains constant despite nonzero variance; X , however, is informative through curvature despite its zero linear coefficient.

3.3 Why the next loop must use actual finite-profile disks

The actual retained finite- q matrix contains different face-flip weights q^4 and q^5 . With

$$p(q) = 2 + 5q + 5q^2 + 6q^3 + 3q^4, \quad c_S(q) = \frac{q^4(1 + q)^2(1 + q^2)}{32p(q)}, \quad (8)$$

the coherent retained center has linear imaginary coefficient $c_S(q)$ in z , approaching $1/84$ but not constant in q . Thus the exact surrogate fibers need not preserve even the finite-profile center. Distinct centers do not by themselves separate full-system predictions: one must include their complete errors. Nor may a value-only bound be differentiated; $e_N(t) = \varepsilon \sin Nt$ has uniformly small values but derivative εN at zero. AI1 therefore closes a scoped identification question and opens a physical discrimination question.

4 AI2: separating two actual finite-profile predictions**

4.1 New observables require a new complete support

Use the free unit cube with origin $(3, 1, 0)$ and the prescribed physical six-cycle X . Let f_4 and f_5 be the opposite yz faces with anchor sums four and five, and let $Y_r = X \triangle f_r$ be the corresponding simple six-cycles. Define actual character multiplication operators, extended by identity,

$$B_r = (\chi_X + \chi_{Y_r})/\sqrt{2}, \quad \psi_r = B_r \Omega, \quad r = 4, 5. \quad (9)$$

Each operator is bounded, real, self-adjoint and gauge-invariant. Conditional Haar integration along the common four-link path and the two independent two-link paths gives mean zero, variance one and fourth moment $5/2$. The exact multiplication norm is $J = 2\sqrt{2}$, not one: its triangle upper bound is approached on positive-measure neighborhoods of identity holonomies. A normalized source vector therefore does not authorize replacing its multiplier by a norm-one rank operator.

Each source uses eight free links, but their union has ten. The complete-factor collar must start from that union. For depths $k = 0, \dots, 5$, independently reconstructed retained-face counts are

$$N_k = (2, 25, 143, 359, 695, 1171). \quad (10)$$

The last collar contains 993 reference factors and 1857 complete links. Copying the earlier eight-link seed would omit part of B_5 . All six cube faces occur by depth one. The inherited AA2 averaged-orbit and exterior-exclusion theorem applies to the entire sixteen-state component containing both sources; it is not an assertion that this component is the entire regional energy shell. The check reconstructs all twenty exterior touching faces and retains their loading and nonzero frequency conditions.

4.2 From connected words to the full scalar disk

Each complete reference factor contains at most ten links, each link meets at most four faces, and an interaction adds at most three factors. Counting every ordered connected word, repetition, factor-two commutator and time simplex yields $(10/3)_n/n!$. For $x = 15z < 1$, a conservative bound on the *whole* omitted series is

$$D_k^{(10)}(x) = \frac{(10/3)_{k+1}}{(k+1)!} \frac{x^{k+1}}{(1-x)^{k+5}}. \quad (11)$$

Taylor's positive integral remainder has exponent $k + 13/3$; rounding it upward to $k + 5$ increases the bound. This accounts for paths leaving the collar and returning. It is neither a finite-spin cutoff nor a global propagator-norm statement.

The retained symmetric 16×16 matrix $Q(q)$ has physical face-flip entries q^4 or q^5 and row norm at most six. Set

$$s_q = \frac{3z(1+q)^2(1+q^2)}{p(q)}, \quad v = s_q/96, \quad h_r = \langle \psi_r, e^{ivQ(q)} \psi_r \rangle. \quad (12)$$

Its exact first moments are $\langle Q \rangle_{\psi_4} = q^4$ and $\langle Q \rangle_{\psi_5} = q^5$; at $q = 1$ both moment lists through degree six equal $(1, 1, 4, 8, 32, 80, 320)$. The computations evaluate rational moments through degree eight. The exponential center error is separately bounded by $(6v)^9/9!$.

The inherited canonical magnetic-profile sum is $b(q) = p(q)/[24(1-q)^3(1+q)^2(1+q^2)]$. Write $\tau_q = \eta/[8b(q)]$, $M_k = N_k/24$ and $\hat{d}_q = \tau_q \sqrt{b(q^2)/96}/[(1-\eta)/8]$. The complete physical-to-center radius is bounded by

$$R_{\text{phys}} = 48\hat{d}_q + 8D_k^{(10)}(15z) + 64\tau_q M_k(1 + 3zM_k), \quad R = R_{\text{phys}} + \frac{(6v)^9}{9!}.^{**} \quad (13)$$

The terms respectively retain the stationary connected-state correction, full spatial tail, both source and vacuum averaging replacements, and arithmetic center error. The constants retain $J^2 = 8$ and $1 + J < 4$. A rational upward square-root enclosure controls $\hat{d}_q$. Fractions determine the verdicts; decimal values are displays. Omitting multiplier loading, connected-state correction or the returned spatial paths would invalidate the disk.

4.3 A successful pairwise test and an unsuccessful ratio certificate

Freeze $z = 10^{-6}$ and compare, for each $u \in \{10^{-12}, 10^{-18}, 10^{-24}\}$,

$$H_1 : (q, \eta) = (1 - u, 1/2), \quad H_2 : (q, \eta) = (1 - 2u, 1/16). \quad (14)$$

Both satisfy $\eta(1-q)^3 = u^3/2$. With common independently fixed α , they therefore have the same carrier and the same physical time

$$T = 2 \cdot 10^{-6} u^{-3} \hbar/\alpha. \quad (15)$$

For each $k \in \{3, 4, 5\}$ and probe r , compare rational imaginary centers. A positive margin

$$\mathfrak{m}_r = \left| \text{Im } h_{r,1}^{(8)} - \text{Im } h_{r,2}^{(8)} \right| - R_1 - R_2 > 0 \quad (16)$$

proves disjoint complex disks and hence distinct actual scalar predictions. Overlap only means this sufficient test failed. It does not establish equality of the two exact models.

Only the cell $u = 10^{-18}$, $k = 5$ gives a positive margin for either probe. The conservative margins are about 3.21452×10^{-26} for B_4 and 4.40500×10^{-26} for B_5 . Equal additional absolute scalar-error radii must each be strictly below half the relevant margin. Its physical time is $2 \times 10^{48} \hbar/\alpha$. This is a conditional mathematical distinction between two named hypotheses, not feasible laboratory calibration or an inverse valid on a continuous parameter domain. Timing uncertainty cannot silently be assigned the same scalar-error allowance without a separate sensitivity bound.

The retained coefficient ratio is exactly $q^5/q^4 = q$. For actual finite-time values y_r , however, spectral calculus first gives

$$|\text{Im } h_r - vq^r| \leq 36v^3, \quad |y_r - vq^r| \leq E_r := R_r + 36v^3. \quad (17)$$

If $vq^4 - E_4 > 0$, interval division yields

$$\frac{vq^5 - E_5}{vq^4 + E_4} \leq \frac{y_5}{y_4} \leq \frac{vq^5 + E_5}{vq^4 - E_4}. \quad (18)$$

If the denominator enclosure contains zero, no ratio inference is made. These simple ratio intervals overlap in all nine cells, including the successful scalar-discrimination cell. Their cubic sine-remainder floor is too conservative for that inverse. The result therefore resolves a pairwise scalar question while preserving the failed ratio objective. No derivative of the full-system correlation is inferred from a value-only disk.

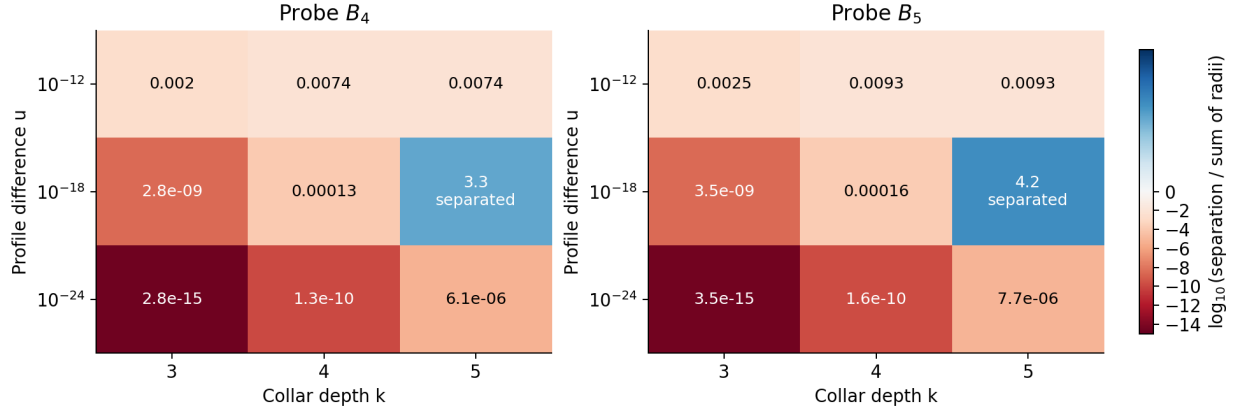


Figure 1: AI2: separation of imaginary centers divided by the sum of complete radii, computed from exact output fractions. Values above one certify disjoint disks. The only successful grid cell is $u = 10^{-18}$, $k = 5$ for both probes. Logarithmic color prevents the large failed range from hiding the threshold; the figure is a display of exact decisions, not a numerical proof.

5 AG1: one selected-source homogeneous correction**

5.1 The actual model and the term still outside this result

After AI2's scoped admission, the advisor selected the homogeneous-update bottleneck. Work in the fixed energy unit $\delta = \alpha/8$ with the homogeneous reference $G = H_0 + \sum_b D_b$ and the actual selected S1 cubic-source family $A = \sum_b A_b$. Each A_b is an identity-extended local mixing term on a complete four-factor star. It is a specified part of a repeated-anchor cubic contribution, not the entire cubic coefficient or transformed remainder. With $M = 7|\tau|$, the available source norm satisfies

$$r_* := \sup_b \|A_b\| \leq r_*^{\text{bd}} = \frac{4}{3} \left(\frac{7}{12} \right)^{3/2} |\tau|^3. \quad (19)$$

Specifically, the inherited source is $A_b = |w_b\rangle\langle\Omega_b| + \text{adjoint}$, where

$$v_b = \phi_b \Omega_b, \quad u_b = (H_{0,Z_b}|_{Q_b})^{-1} v_b, \quad w_b = -\langle u_b, v_b \rangle u_b - \frac{\|u_b\|^2}{3} v_b. \quad (20)$$

The cubic expression in (19) is an upper bound, not an evaluated equality. The frozen main interval is $0 \leq M \leq 10^{-3}$, with a predeclared refinement on $0 \leq M \leq 10^{-4}$. The auxiliary support-cardinality weight is $\rho = 9/4$, while the required output uses the original weight two. These are proof parameters within the same homogeneous family; the filter time $T = 3$ is not the canonical readout clock.

Translation of identical complete blocks gives the same actual norm $r = \|A_b\| > 0$ for every retained star when $\tau \neq 0$. Deleting boundary stars does not modify a retained source. If m_Λ is the maximum number of retained source stars containing a root, then $1 \leq m_\Lambda \leq 4$ for a nonempty family and its declared indexed norm is exactly

$$A_a := \|\{A_b\}\|_a = m_\Lambda a^4 r, \quad \frac{A_\rho}{A_2} = \left(\frac{9}{8} \right)^4. \quad (21)$$

This is a genuine input denominator even though r is not numerically evaluated. Substituting its upper bound into a denominator would be invalid. Empty families and $\tau = 0$ have zero update and are treated without division. Physical energies multiply these dimensionless bounds by δ , with the original positive spacing and physical reference scales fixed.

If the full previously transformed Hamiltonian contains other terms E , the correct target remains $G + A + E$. Controlling $G + A$ does not control E . The exact equation below retains its transport

and states the needed conditional norm premise. In particular AG1 is not a completed correction of every term in the original homogeneous Yang–Mills Hamiltonian.

5.2 Domain-safe conjugation and the full nonlinear remainder

For the triangular kernel $p_T(t) = (1 - |t|/T)_+/T$ and its odd integrated primitive $h_T(t) = \text{sgn}(t)(1 - |t|/T)^2/2$ on $0 < |t| < T$, let

$$\mathcal{R}_T(A_b) = \int p_T(t) e^{itG} A_b e^{-itG} dt, \quad \mathcal{J}_T(A_b) = -i \int h_T(t) e^{itG} A_b e^{-itG} dt. \quad (22)$$

Their basic kernel identities are $h'_T = \delta_0 - p_T$, $\int p_T = 1$ and $\int |h_T| = T/3$. Weak integration by parts gives a bounded commutator $[\mathcal{J}_T(A_b), G] = -A_b + \mathcal{R}_T(A_b)$. The inherited adjoint-domain argument then gives graph-domain preservation. In a finite containing cuboid the sum $S = \sum_b \mathcal{J}_T(A_b)$ is bounded and skew-adjoint, preserves $D(G)$, and obeys $[S, G] = -A + R$ with $R = \sum_b \mathcal{R}_T(A_b)$.

On that domain the exact update is

$$e^S(G + A)e^{-S} = G + R + N, \quad N = \sum_{n \geq 1} \frac{\text{ad}_S^n(nA + R)}{(n+1)!} = \int_0^1 e^{s \text{ad}_S} (s[S, A] + (1-s)[S, R]) ds. \quad (23)$$

This identity expands the first commutator of G only after replacing it by the bounded $-A + R$; all subsequent terms involve bounded interactions. To derive the coefficient, combine $\text{ad}_S^n A/n!$ from the bounded A conjugation with $\text{ad}_S^n(-A + R)/(n+1)!$ from G . The result is $\text{ad}_S^n(nA + R)/(n+1)!$. Thus the leading nonlinear term is $[S, A + R]/2$. A G -only BCH formula would omit a required contribution. The resonant filter residual is retained: the Fourier multiplier $\text{sinc}^2(\omega T/2)$ equals one at zero frequency.

5.3 Why the full commutator series needs weight loss

For an indexed interaction with complete connected supports, define

$$\|\Phi\|_a = \sup_x \sum_{X \ni x} a^{|X|} \|\Phi_X\|, \quad a \geq 2. \quad (24)$$

Repeated terms with the same support retain their indices unless explicitly grouped. The forward complete rooted count starts at $N_0(a) = A_a$. Adding one interaction to a depth- n connected word costs $2Ma^3$; the two root locations count stars meeting the existing union and incoming stars containing the root. This gives

$$N_{n+1}(a) \leq 8Ma^3(8 + 3n)N_n(a), \quad N_n(a) \leq A_a(24Ma^3)^n(8/3)_n. \quad (25)$$

All ordered repetitions, incoming overlaps and complete supports remain. Finite boundaries only delete terms. The time simplex and exact kernel moments then give, with $x_a = 72a^3M$ and $F_a = (1 - x_a)^{-3}$,

$$\begin{aligned} \|R\|_a &\leq A_a \sum_{n \geq 0} \frac{2(8/3)_n x_a^n}{(n+2)!} \leq A_a F_a, \\ \|S\|_a &\leq A_a \sum_{n \geq 0} \frac{6(8/3)_n x_a^n}{(n+3)!} \leq A_a F_a. \end{aligned} \quad (26)$$

The last inequalities bound the entire positive orbital series of exponent $8/3$ by exponent three, using the unit absolute masses at $T = 3$. At the main endpoint $x_\rho = 6561/8000 < 1$, so these

estimates hold continuously over the whole interval. The twelve interior crossings also preserve the inherited per-source residual bound $(24M + 2/3)/(1 - M) \leq 2072/2997$; that alone does not prove weighted-family contraction.

For $a > b \geq 2$, overlap counting in a commutator has two possible root locations. After fixing one support, the overlapping second supports cost its cardinality. Nonempty overlap supplies a factor $b^{-1} \leq 1/2$, and $\sup_{m \geq 1} m(b/a)^m \leq 1/[e \log(a/b)]$ absorbs that cardinality. Therefore

$$\|[\Phi, \Psi]\|_b \leq \frac{2}{e \log(a/b)} \|\Phi\|_a \|\Psi\|_a. \quad (27)$$

Split the loss $\log(\rho/2)$ equally over n commutators. Using $n! \geq (n/e)^n$ gives

$$\frac{\|\text{ad}_S^n X\|_2}{n!} \leq \vartheta^n \|X\|_\rho, \quad \vartheta = \frac{2 \|S\|_\rho}{\log(\rho/2)} \leq 18 \|S\|_\rho, \quad (28)$$

where $\log(9/8) \geq 1/9$ gives the rational majorant. If a complete translated-source estimate proves a chosen upper bound $\bar{\vartheta} < 1$, summing every BCH term yields

$$\|N\|_2 \leq \frac{\bar{\vartheta}}{1 - \bar{\vartheta}} \left(\|A\|_\rho + \frac{1}{2} \|R\|_\rho \right). \quad (29)$$

The complete rooted estimates, including repeated words, incoming overlaps and twelve possible interior crossings, are needed to justify the finite input norms. A finite-volume operator bound growing with $\|S_\Lambda\|$ is not a substitute for a volume-independent interaction bound. Nor can the same positive weight margin be spent indefinitely without a new summable-loss induction.

For additional terms with a separately justified premise $\|E\|_\rho \leq e_*$,

$$e^S(G + A + E)e^{-S} = G + R + N + e^S E e^{-S}, \quad \|e^S E e^{-S}\|_2 \leq \frac{e_*}{1 - \bar{\vartheta}}. \quad (30)$$

The parameter e_* is a conditional error inventory, not a fitted physical term. This inequality does not establish that the actual omitted E satisfies it.

5.4 The one-step estimates and the narrower contraction

The forward rational majorant uses $r \leq M^3/567$ and $\bar{\vartheta} = 72\rho^4 F_\rho M^3/567$. At $M = 10^{-3}$ it gives $\bar{\vartheta} < 5.5920 \times 10^{-7}$ and

$$\frac{\|N\|_2}{A_2} \leq \nu(M) := \frac{\bar{\vartheta}}{1 - \bar{\vartheta}} \left(\frac{9}{8} \right)^4 \left(1 + \frac{F_\rho}{2} \right) < 7.7851 \times 10^{-5}. \quad (31)$$

The underlying fractions, not the displayed rounding, establish the bound. Every positive factor increases with M before its pole, so endpoint inequalities cover the continuous interval. This controls the whole nonlinear remainder, while the loose main residual majorant does not establish contraction there.

For $0 < M \leq 10^{-4}$, split R into its exact term without a D_b interaction and the complete positive tail containing at least one D_b . On the original star the first part is $|f_3(H_{0,Z_b})w_b\rangle\langle\Omega_b| + \text{adjoint}$, where $f_3(E) = \text{sinc}^2(3E/2)$. The inherited source is orthogonal to the reference vacuum and its free local Hamiltonian has gap at least one. Thus $|f_3(E)| \leq 4/(9E^2) \leq 4/9$ on that source. The $n \geq 1$ part of (26) costs at most $A_2(F_2 - 1)$. Consequently

$$\frac{\|R + N\|_2}{A_2} \leq \frac{4}{9} + (1 - 576M)^{-3} - 1 + \nu(M) < 0.639242 < \frac{2}{3}^{**}. \quad (32)$$

This contracts against the actual chosen input norm (21), not two upper budgets. The independently derived reverse count uses different positive word majorants and gives the weaker bound

0.923603 on the same interval. Post-exchange independent review verifies the stronger rooted recurrence used here. Unequal conservative constants are not conflicting evaluations of an exact observable. The narrower estimate is a predeclared refinement within AG1, not a fourth loop.

The skeptic separately retained more of the exact filter integration. With $\mathcal{F}(x) = (4-x)/(1-x)^2$, the positive $n \geq 1$ tail has coefficient $t(M) = \frac{16}{3}[\mathcal{F}(624M)-4]$ multiplying the actual r_* . The bound follows from the exact integrated coefficient $32(4+3n)x^n/[(n+1)(n+2)]$ and $(n+1)(n+2) \geq 6$. Since the actual input norm is at least $16r_*$, combining this tail with the skeptic's full nonlinear coefficient $n_{\text{sk}}(M)$ gives

$$\frac{\|A'_{\text{new}}\|_2}{\|A\|_2} \leq \frac{4}{9} + \frac{t(M) + n_{\text{sk}}(M)}{16} < 0.605 \quad (0 < M \leq 10^{-4}). \quad (33)$$

This reviewer refinement, independently frozen and then cross-checked, improves a sufficient bound to about 0.604164. The conservative reviewed headline remains one-step selected-source contraction below $2/3$. It is neither an exact dynamical contraction factor nor an extra physics loop.

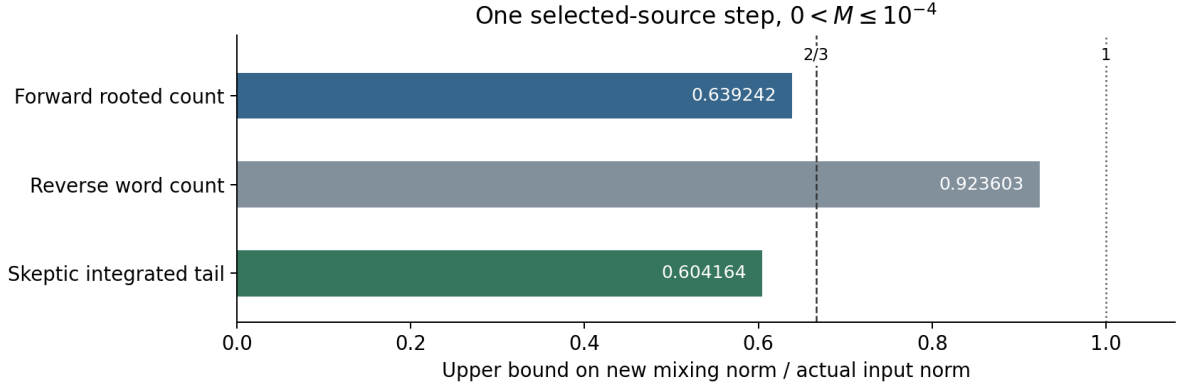


Figure 2: AG1 exact-output-backed sufficient bounds on the new selected mixing family, relative to the actual input norm. The producer bounds also cover $R + N$ before projection. Different positive-tail estimates produce different constants; none is an exact contraction measurement. The $2/3$ headline is supported by the reviewed forward and skeptic estimates.

For each generated self-adjoint indexed term K_X of $K = R + N$, keep its full support and define $P_X = |\Omega_X\rangle\langle\Omega_X|$, $Q_X = I - P_X$ and

$$\begin{aligned} c_X &= \langle\Omega_X, K_X \Omega_X\rangle, & A'_X &= P_X K_X Q_X + Q_X K_X P_X, \\ D'_X &= Q_X (K_X - c_X I) Q_X, & K_X &= c_X I + A'_X + D'_X. \end{aligned} \quad (34)$$

The costs are $|c_X| \leq \|K_X\|$, $\|A'_X\| \leq \|K_X\|$ and $\|D'_X\| \leq 2\|K_X\|$, hence at most four times the budget when all three are summed. The scalar keeps its declared support for a density budget and may sum extensively. It is a reference expectation, not an interacting ground energy. “Diagonal” means the reference vacuum/complement split, not all excited spectral blocks. The new mixing family alone inherits (32); the changed diagonal, scalar and conditional E cannot be discarded. The consumed weight margin, new supports and changed reference obstruct immediate iteration.

The checkers include both coupling signs, zero/empty cases, boundary multiplicities, all twelve crossings, complete positive tails, omitted-residual and wrong-BCH controls, scalar reconstruction and conditional E transport. Independent finite rational matrix fixtures test algebraic coefficients; they are diagnostics, not the physical completeness proof. The all-order support and graph-domain arguments carry the result. The full original remainder inventory, all-stage induction, homogeneous numerical spectral gap, physical matching and continuum construction remain open.

A Contributions and remaining proof obligations

Contribution	Earlier understanding → present change	Possible application	Limitation and next missing step
AI1 fibers**	Nontrivial endpoint formulas → exact classification of the combinations identified by each declared observation convention.	Remove redundant parameter fits; choose which calibration must be independent.	Surrogate identities are not finite- q model equivalences. Prove and use actual finite-profile error disks.
AI1 carrier**	Demodulated slow signal → rank-two identification of carrier and slow rates with a residual η - q fiber and explicit sampling aliases.	Plan phase-aware, nonaliased observation protocols.	Requires calibrated physical time/action/phase. No implemented clock or derivative transfer theorem.
AI2 probes and support**	One coherent face type → two actual multiplier sources with weights q^4, q^5 , ten-factor common seed and recomputed complete collars.	Design observables sensitive to profile differences; prevent omitted-channel understatement.	Source-specific canonical family. The component is not the whole regional energy shell.
AI2 full-scalar test**	Retained-center differences → disjoint complete scalar disks for one frozen grid cell. Other cells and all simple ratio tests remain insufficient.	Reject one of two named model hypotheses when the full error allowance is met.	$T = 2 \times 10^{48} \hbar / \alpha$ and roughly 10^{-26} scalar margins; no feasibility claim or continuous inverse.
AG1 one update**	Individual-source estimates → an entire nonlinear remainder bound and actual selected-source contraction below $2/3$ on the narrower interval.	Identify the terms, input denominator and norm costs an iteration must control.	Restricted $G + A$ source sector and one spent weight margin. Other original terms require explicit transport; no all-stage or complete-Hamiltonian contraction follows.

The two-star notation marks workbench specialization, derivation or implementation, not established scientific priority. These results may be useful even when a desired certificate fails: a failed scalar separation or contraction upper bound identifies a limitation of the sufficient estimate. It does not prove equality, divergence or impossibility of the underlying dynamics.

A.1 Why a percentage of solving Yang–Mills is not estimable

The three requested investigations have final scoped gates: the research task count is 3/3. That known denominator does not supply a percentage of a mathematical proof. The expert lenses and skeptic agree that no defensible numerical fraction of the Yang–Mills solution is identifiable from pages, loops, equations, network nodes or finite-model successes. The remaining obligations depend on each other and have no justified equal weights or known total amount of work.

The official problem asks for a nontrivial four-dimensional quantum Yang–Mills theory with the required axiomatic strength and a positive mass gap for every compact simple gauge group. The official status remains unsolved at the recorded source check. The workbench’s outstanding obligations are:

Obligation	Missing implication after Round27
Homogeneous control and matching	Convergent changed-source/changed-diagonal induction and a proved map on states, observables, domains and generators.
Uniform limiting estimates	Bounds surviving growing volume and regulator removal along a physically matched trajectory in the required topology.
Nontrivial continuum construction	The requested four-dimensional limiting theory and nontrivial local fields.
Axiomatic properties	Limiting locality, covariance, positivity/reconstruction and the required ultraviolet behavior.
Positive physical gap	A positive finite threshold in the actual constructed limiting theory, in fixed physical units.
Gauge-group coverage	The required compact simple groups beyond the workbench's $SU(2)$ emphasis.

For a dimensionless cutoff operator $L(a)$ and energy convention $E(a)$, the identity $\Delta_{\text{phys}}(a) = E(a)\Delta_L(a)$ only converts units. A positive finite-cutoff $\Delta_L(a)$ proves no positive limiting physical gap without a justified scale, limiting construction and uniform estimate. An unknown completion percentage is not a claim of zero research value, nor an estimate of remaining effort.

The panel's next unexecuted priority is AG2: recover the original transformed Hamiltonian at the source checkpoint, inventory every remaining term in E , prove its stronger support-norm budget, and test whether its transported mixing cost fits inside the selected-source contraction margin. Only then should a changed-diagonal step be frozen with its own domain and remaining weight hypotheses. A cancellation-aware AI3 finite-value inverse remains an alternative observable branch. Neither is executed in this three-loop continuation.

B Reproduction, review and linked sources

The mathematical reports remain authoritative for their frozen assumptions, complete-support definitions and detailed controls. This manuscript summarizes them without rewriting the accepted gates or Draft01. Every displayed plot is reconstructed from exact JSON fractions; converting to floating point occurs only after the exact decision, for rendering. This is not a new physical simulation.

From the repository root, replay all three producer pairs into a fresh absolute directory outside the repository:

```
python3 -B research/round27/reproduce.py \
  --output /absolute/new/round27-replay
python3 -B research/round27/reproduce.py --optimized \
  --output /absolute/new/round27-optimized
python3 -B papers/round27-addendum/build.py
```

The first two commands verify source/snapshot bindings and byte-identical fresh outputs. Output directories must not already exist. Optimized mode checks that Python optimization does not disable the controls. They do not constitute formal verification of an operator proof or independent empirical observations. The final command builds the plot and the PDF with the installed Python scientific stack, ordinary L^AT_EX and Poppler; no dependency download is needed in the authoring environment. It requires all final scoped gates.

The addendum's `manifest.json` binds its source files, final PDF and plots, together with the source gates/reports it summarizes. `pdf-qa.json` records the final PDF hash, page count, text/layout diagnostics and the separate visual review. Temporary page rasters and TeX products are excluded from the source package. The PDF is rendered and visually checked after the final content change. Rebuilding requires repeating that visual review; successful compilation alone is insufficient.

Repository evidence. Within `research/round27/`, see `contracts/`, the separate `forward/` and `reverse/` reports and exact outputs, `skeptic/`, and `advisor/`. The round's `SOURCES.md` summarizes the survey; each of the five `experts/` directories contains a report and machine-readable inventory with source roles and reading limits. The earlier manuscript is preserved in `papers/draft-01/`. The public repository is [occult-kranti/yang_mills_workbench](https://github.com/occult-kranti/yang_mills_workbench).

Selected primary links and actual reading scope

These pointers summarize the expert records; the full inventory retains all 55 entries, inaccessible endpoints and discussion links. No full copyrighted book text is redistributed. Source findings in this addendum are attributed to the recorded panel readings, not to an independent new audit of every linked work.

- [1] Newton, *Opticks*, Query 31 and General Scholium. Selected closing methodological and speculative passages. *Experiments*, Add.3973: selected dated operations and residues; *Keynes MS.28*: English transcription, not a full Latin commentary audit.
- [2] Tesla, [US787412A](#) and [Colorado Springs entry](#), 7 December 1899. Mechanism description and one tuning table. [FBI institutional account](#): custody statement; the large catalogue scan was not read.
- [3] Jung, *Psychology and Alchemy* and *Synchronicity*. Publisher-hosted chapter openings only. Zabriskie, [introduction to Atom and Archetype](#): context and excerpts, not the full correspondence.
- [4] Penrose, *Angular momentum: an approach to combinatorial space-time*. Local admissibility and possible zero norm; [handwritten notes](#): first two pages visually inspected. Gravitational reduction was treated separately and at abstract depth.
- [5] Feynman, *The Qualitative Behavior of Yang-Mills Theory in 2+1 Dimensions*. Primary abstract and bibliographic scope; full text not retrieved. *Cargo Cult Science* supplies methodological context.
- [6] Jaffe–Witten, *Quantum Yang–Mills Theory*; official [Clay status page](#). Target statement and selected limit obligations; status checked 22 September 2026.
- [7] Ciavarella, Burbano and Bauer, [arXiv:2503.11888v2](#): selected local truncation and large- N sections. Cao, Nissim and Sheffield, [arXiv:2505.16585v3](#): introduction and Theorem 1.2 context, not a complete proof audit.
- [8] Bonanno et al., [arXiv:2511.07355v1](#): selected scale, topology and volume sections. Froland and Grabowska, [arXiv:2606.14842v1](#): selected apparatus, estimator and digitization discussion.
- [9] Chevyrev and Shen, [arXiv:2503.03060v2](#): inspected theorem and explicit remaining stochastic three-dimensional construction limits. No four-dimensional pure-theory conclusion is imported.
- [10] Kane, Hariprakash and Bauer, [arXiv:2506.16559v1](#); Negro et al., [arXiv:2606.20029v1](#); [arXiv:2609.07886v1](#). Selected continuum-error, step-scaling and digitization comparisons in their stated models.
- [11] Petrica and Popescu, [arXiv:2408.14616v2](#): local-inverse theorem and proof inspected; its finite-dimensional hypotheses are not automatically imported into this model. Maclaren and Nicholson, [arXiv:1904.02826v4](#): uniqueness versus stable estimability.
- [12] Nachtergaele, Sims and Young, [arXiv:1810.02428](#): abstract and bibliographic scope checked as a quasi-locality comparison. It supplies no unverified estimate for our support norm.